CGT 16400 – Graphics for Construction & Civil Engineering

Public-Facing Syllabus – Spring 2026

Course Number & Title

CGT 16400 – Graphics for Construction & Civil Engineering

Credit Hours: 2

Instructional Modality

Face-to-Face

Instructor

instructor(s):	Clark Cory PhD
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Course Description

This introductory course focuses on construction and civil engineering graphics used in professional practice. Students develop foundational skills in spatial visualization, sketching, computer-aided design (CAD), and introductory modeling to communicate construction information clearly and accurately using accepted industry standards.

Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. **Understand the Role of Construction Graphics**: Demonstrate knowledge of the history, theories, and importance of engineering design processes and graphics in construction, including their role in real-world problem-solving and communication.
2. **Apply Geometric and Graphic Standards**: Develop proficiency in creating and solving 2D and 3D geometric forms and applying construction graphic standards and codes to engineering problems.
3. **Utilize Computer Graphics and Visualization**: Effectively use computer-aided design (CAD) tools and 3D visualization skills to analyze, solve, and communicate construction challenges.
4. **Enhance Communication Through Graphic Conventions**: Present solutions clearly and accurately using appropriate 2D and 3D graphical communication standards and conventions.
5. **Foster Creative Problem-Solving Skills**: Develop technical skills, creativity, and personal discipline to solve engineering and construction problems using sketching, visualization, and database management strategies.